

Module 2 - Data Modeling and ELT Concepts

Run dbt Projects and verify outputs in the database

1. Verify the orders table.

```
SELECT * FROM demo_dag.customers;  
SELECT * FROM demo_dag.orders;
```

2. Join the orders table with the customers table on the customer_id, then add a WHERE clause to restrict the results to rows where the customer's country is 'USA'.

```
SELECT o.*  
FROM demo_dag.orders o  
JOIN demo_dag.customers c  
  ON o.customer_id = c.customer_id  
WHERE c.customer_country = 'USA';
```

3. Sort the orders by their total value by executing the following query.

```
SELECT  
  order_id,  
  quantity * price AS total_value  
FROM demo_dag.orders  
ORDER BY total_value DESC;
```



4. Perform a simple aggregation by running the following query.

```
SELECT SUM(quantity * price) AS total_revenue  
FROM demo_dag.orders;
```

5. Look at how many orders come from each customer country.

```
SELECT
    c.customer_country AS country,
    COUNT(o.order_id) AS total_orders
FROM demo_dag.orders o
JOIN demo_dag.customers c
    ON o.customer_id = c.customer_id
GROUP BY c.customer_country
ORDER BY total_orders DESC;
```

6. Identify the highest-spending customers.

```
SELECT
    c.customer_name,
    SUM(o.quantity * o.price) AS total_spent
FROM demo_dag.orders o
JOIN demo_dag.customers c
    ON o.customer_id = c.customer_id
GROUP BY c.customer_name
ORDER BY total_spent DESC;
```

